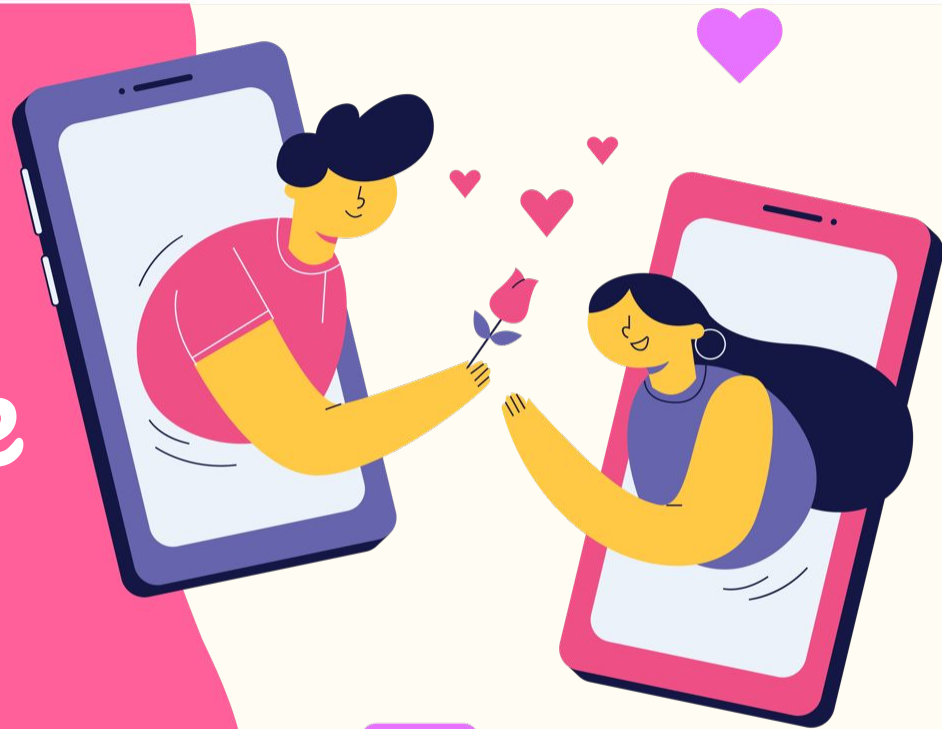


The Date-a-Base

Group 8 - The Naive Babes

Calista Agmata, Katharine Baker, Juliet
McMillan, Grace Pascarella, Rebecca Trailor



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The background is a solid pink color. In the top-left corner, there is a dark red circular shape containing several smaller, lighter red circles of varying sizes. In the bottom-right corner, there is a large dark red circular shape containing several smaller, lighter red circles of varying sizes. A small, solid red circle is located in the middle-right area of the slide.

Motivations & Introduction

Motivation and Introduction



Why this topic

- ★ Online dating is a huge part of modern relationships
- ★ Language is a powerful window into identity
- ★ Address safety concerns of users lying about their age



Why it's interesting to others

- ★ Most people don't realize how much their writing reveals
- ★ Useful for dating platforms, marketers, and researchers
- ★ Connects language, psychology, and data science

Motivation and Introduction



What we hope to learn

- ★ Can age be predicted from writing alone?
- ★ What linguistic features correlate with age?
- ★ How consistent are these patterns across platforms?



What we will show you

- ★ Our age-predictive models trained on OkCupid dating profiles
- ★ Key features used
- ★ Performance results
- ★ Insights into how different age groups write



Data

Original Data



okcupid

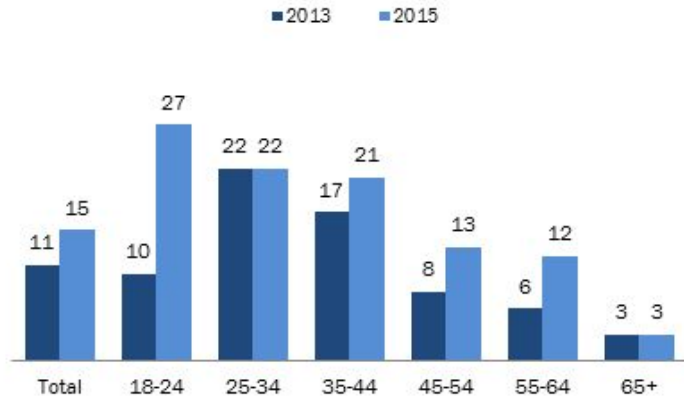
- ★ OkCupid dataset from Kaggle, contains profile information from 60,000 users including age, sex, orientation, and text data from open ended descriptions

Age	Status	Gender	Essay 1	Essay 2	Essay 3
24	single	female	oh goodness. at the moment i have 4 jobs, so it'd be nice to find one i could settle into. other tha...	i'm freakishly blonde and have the same name as that hurricane from a while back.	sports/my softball glove coffee. because nobody likes zombies. kindle loud music/concerts candleligh..
39	single	female	i'm usually pretty exhausted by friday. so usually it's a whole lot of nothin'!	you tell me:)	obscure dances from the '30's and '40's, laughing at myself, being playful in general, and working w...

Who Uses Dating Apps and Why?

Use of online dating sites or mobile apps by young adults has nearly tripled since 2013

% in each age group who have ever used an online dating site and/or mobile dating app



Source: Survey conducted June 10-July 12, 2015.

PEW RESEARCH CENTER

<https://www.pewresearch.org/internet/2016/02/11/15-percent-of-american-adults-have-used-online-dating-sites-or-mobile-dating-apps/>

Why do people use dating apps?



<https://www.biztechcs.com/blog/how-to-make-app-like-tinder/>

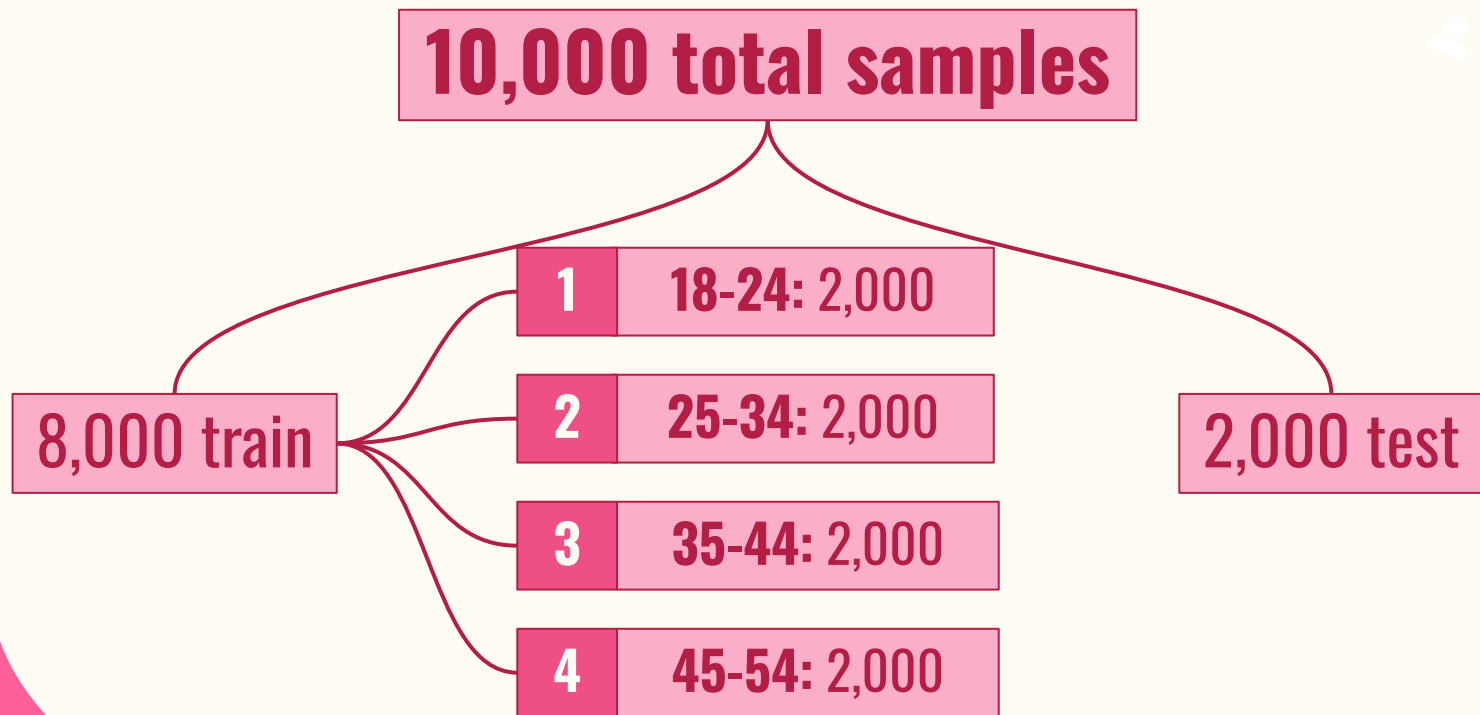
Data Changes

- ★ Only consider **age** and **essay** responses for each entry
- ★ Converted 9 essay responses into one string for processing
- ★ Tokenization: ~300 tokens per response
- ★ Divided set into training (80%) and testing (20%) partitions
- ★ Randomly selected 2000 entries for each age group

Age Groups:

1	18-24
2	25-34
3	35-44
4	45-54

Processed Data



Methods

Goal

Sample input:

“i hated cleaning the
litter box so much, i
taught my cats to use
the toilet.”

Model

Age Groups:

1 18-24

2 25-34

3 35-44

4 45-54

Correct Age!

Baselines

Goal: predict approximate age of user given their essay responses

Stratified: 0.2595

Random: 0.2405

Majority Class: 0.2500

Calculated with scikit-learn metrics

Methods

Goal: predict approximate age of user given their essay responses

1

Mistral AI

- ★ Content based approach
- ★ Zero and multi-shot learning
- ★ 200 testing examples

2

Bag of Words

- ★ Part-of-speech based approach
- ★ 2000 testing examples

3

Word2Vec

- ★ Word embeddings
- ★ SVM to make predictions
- ★ 2000 testing examples

4

tf-idf

- ★ Statistical approach
- ★ 2000 testing examples

5

BERT

- ★ Label encoding
- ★ 2000 testing samples

Approach 1: Mistral AI

- ★ Use **mistral library** from Mistral Ai
- ★ Began with implementing zero-shot learning
- ★ Baseline for comparison with our few-shot approach
- ★ 200 trials with unique inputs from test data

Prompt: “You are able to determine a person's age group based on their writing style. Based on the following text, which age group (18-24, 25-34, 35-44, or 45-54) does the author most likely belong to?
TEXT: {text[:500]}
Please analyze the writing style, references, life experiences, and concerns mentioned. Answer with JUST the age range (18-24, 25-34, 35-44, or 45-54) and nothing else.”

Approach 1: Mistral AI

- ★ Two-shot learning
- ★ 200 trials with unique inputs from test data

Prompt: "You are able to determine a person's age group based on their writing style. Based on the following text, which age group (18-24, 25-34, 35-44, or 45-54) does the author most likely belong to?"

TEXT: {text[:500]}

Here are some examples:

Age 24: oh goodness. at the moment i have 4 jobs, so it'd be nice to find one i could settle into.

Age 39: i'm usually pretty exhausted by friday. so usually it's a whole lot of nothin'!

Please analyze the writing style, references, life experiences, and concerns mentioned. Answer with JUST the age range (18-24, 25-34, 35-44, or 45-54) and nothing else."

Approach 2: Bag of Words PoS



★ Developed BoW using a manual approach

- Preprocessing:
 - Keep stop words
 - `nltk.word_tokenize`, `nltk.pos_tag`
- PoS Frequencies
 - `nltk.FreqDist`
- BoW Arrays

★ Training

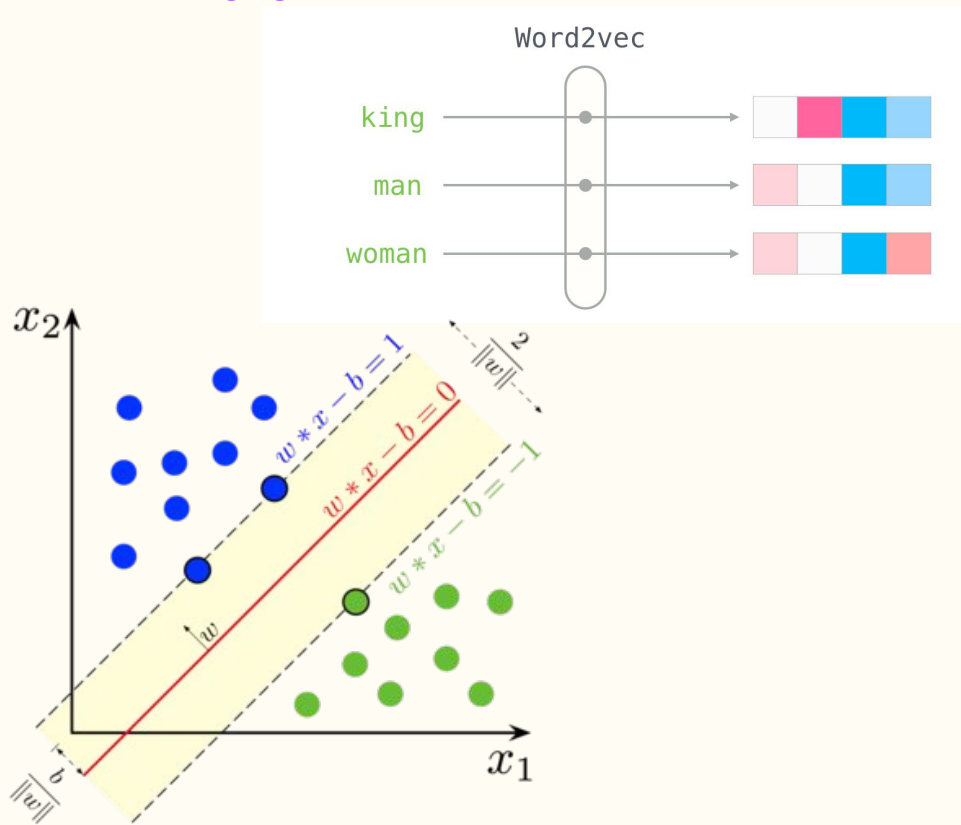
- Gaussian Naive Bayes
 - `sklearn`

Approach 2: Bag of Words PoS

Example BoW PoS Arrays:

PoS	NN	JJ	IN	DT	NNS	VBP	RB	CC	VB	VBG
User x	1	1	1	1	1	1	1	1	0	1
User y	1	1	1	1	1	1	1	1	1	1
User z	0	0	0	0	0	0	0	0	0	0

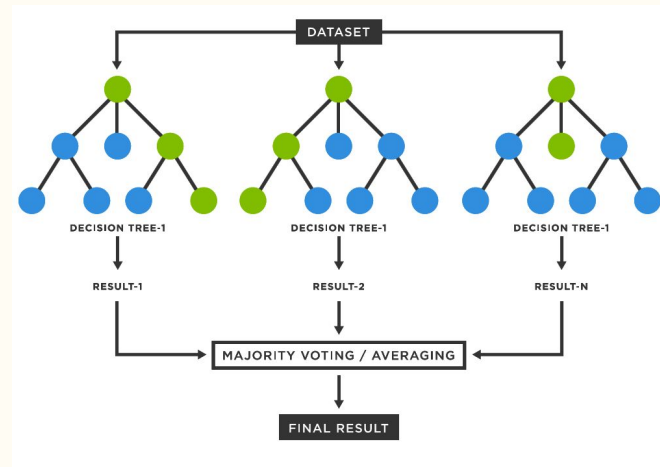
Approach 3: Word2Vec with SVM



- ★ **Preprocessing**
 - Sentence tokenize each essay in training data (**nlk.sent_tokenize**)
 - Word tokenize each sentence (**nlk.word_tokenize**)
 - Append tokens to array containing all tokens
- ★ **Build model based on array (**Word2Vec**)**
- ★ **Create word embedding, averaged feature vectors (**np.array**)**
- ★ **Train with SVM (**sklearn**)**

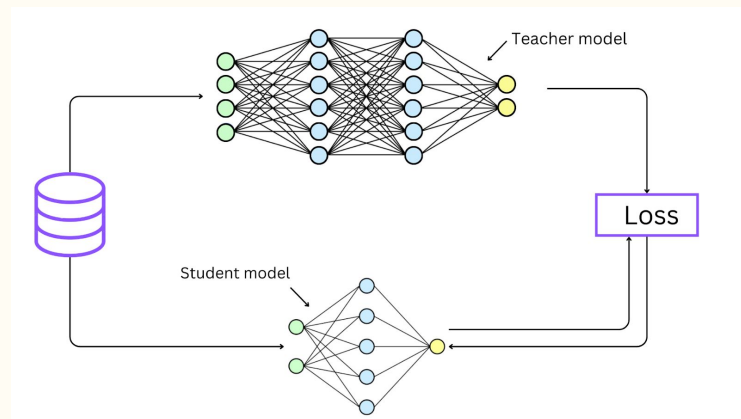
Approach 4: TF-IDF

- ★ Used **TfidfVectorizer** from **sklearn**
 - Limited vocab size to top 5,000 terms
 - Ignored terms occurring in fewer than 5 documents
 - Included unigrams and bigrams
- ★ Trained model with **RandomForestClassifier** to learn patterns from TF-IDF features
- ★ 2000 testing examples



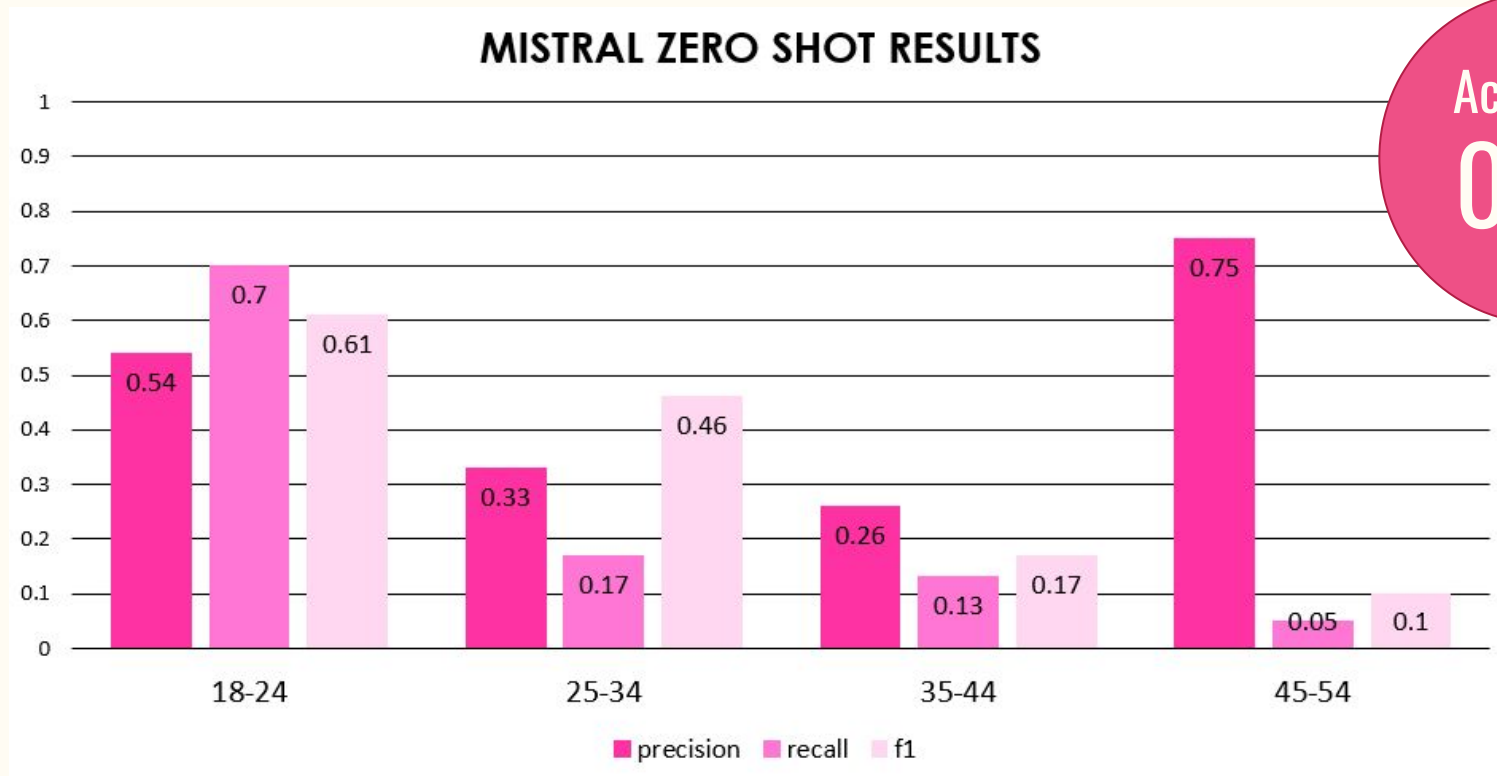
Approach 5: BERT

- ★ Used DistilBERT from HuggingFace for efficiency
- ★ Used **DistilBertForSequenceClassification** model
- ★ Label encoding of age groups using **LabelEncoder**
- ★ Then tokenized essay responses using **DistilBertTokenizer**
- ★ Create a custom **PyTorch** dataset to represent both training and testing essays
- ★ Fine tune BERT with training data
- ★ Applied to testing data

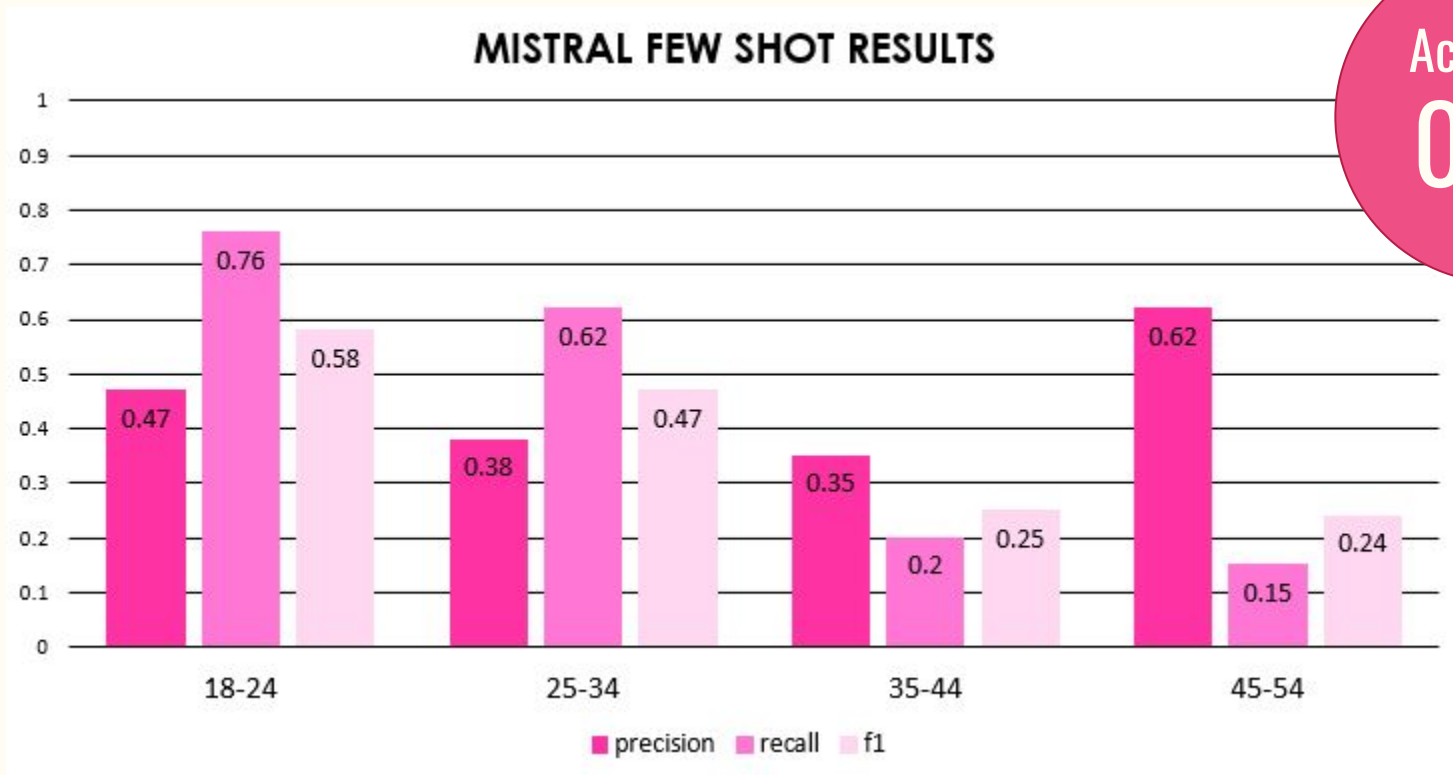


Results & Analysis

Approach 1 (Zero Shot): Results



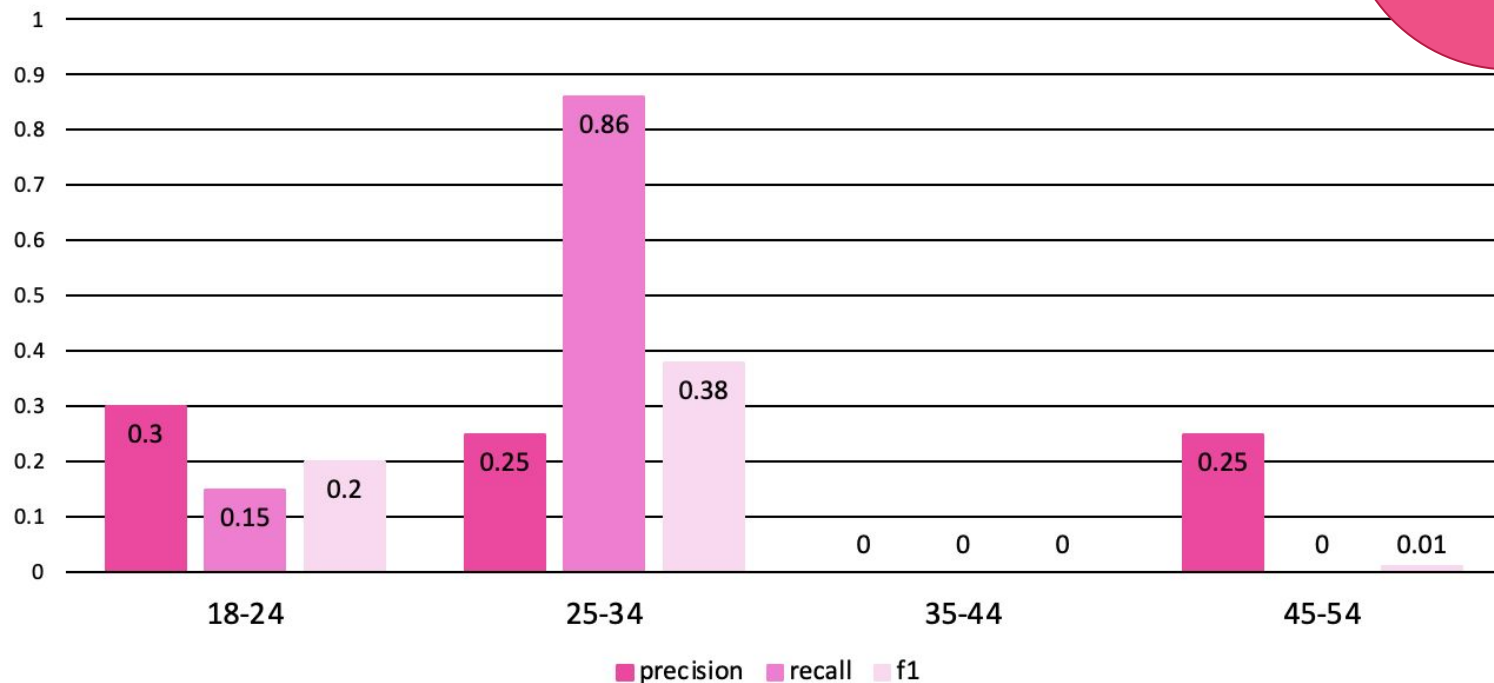
Approach 1 (Few Shot): Results



Approach 2: Results

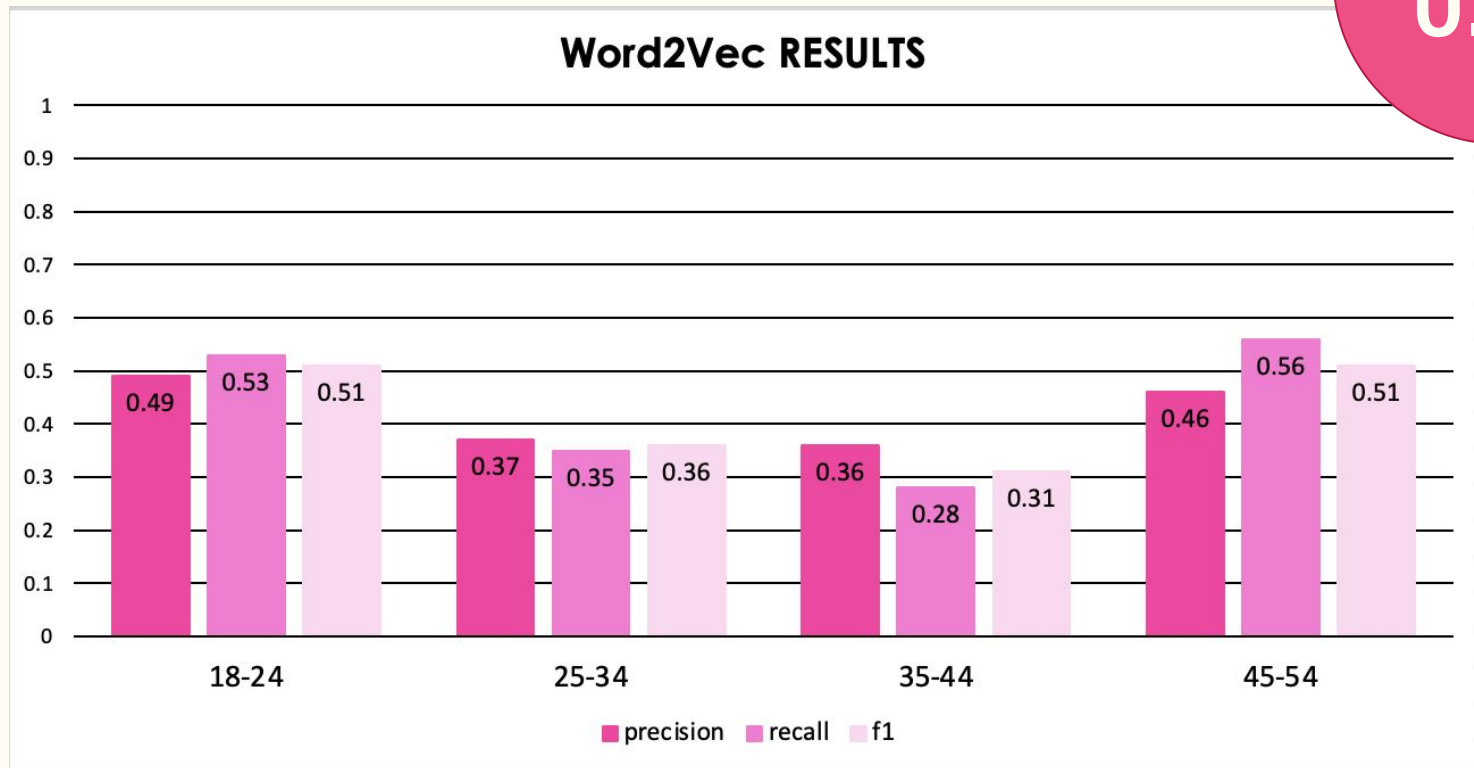
Accuracy:
0.25

BAG of PoS RESULTS



Approach 3: Results

Accuracy:
0.43

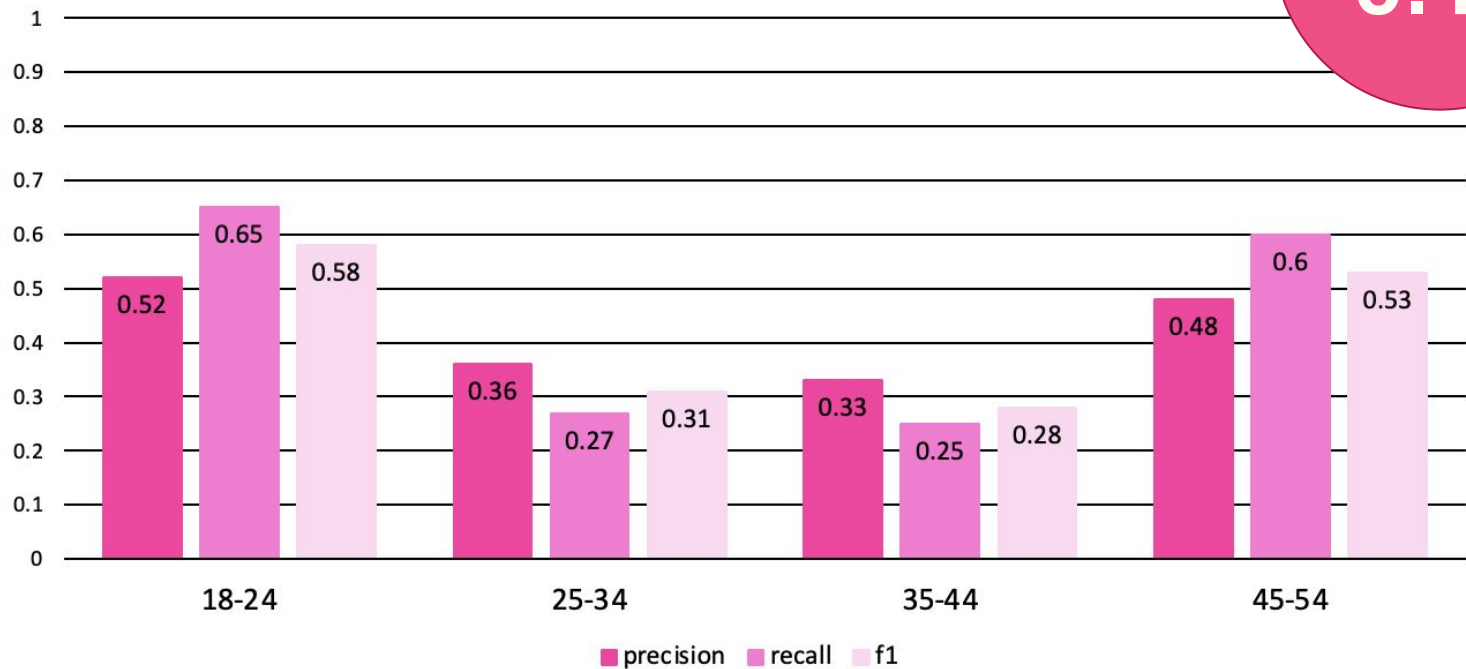


Approach 4: Results

Accuracy:

0.44

TF-IDF RESULTS

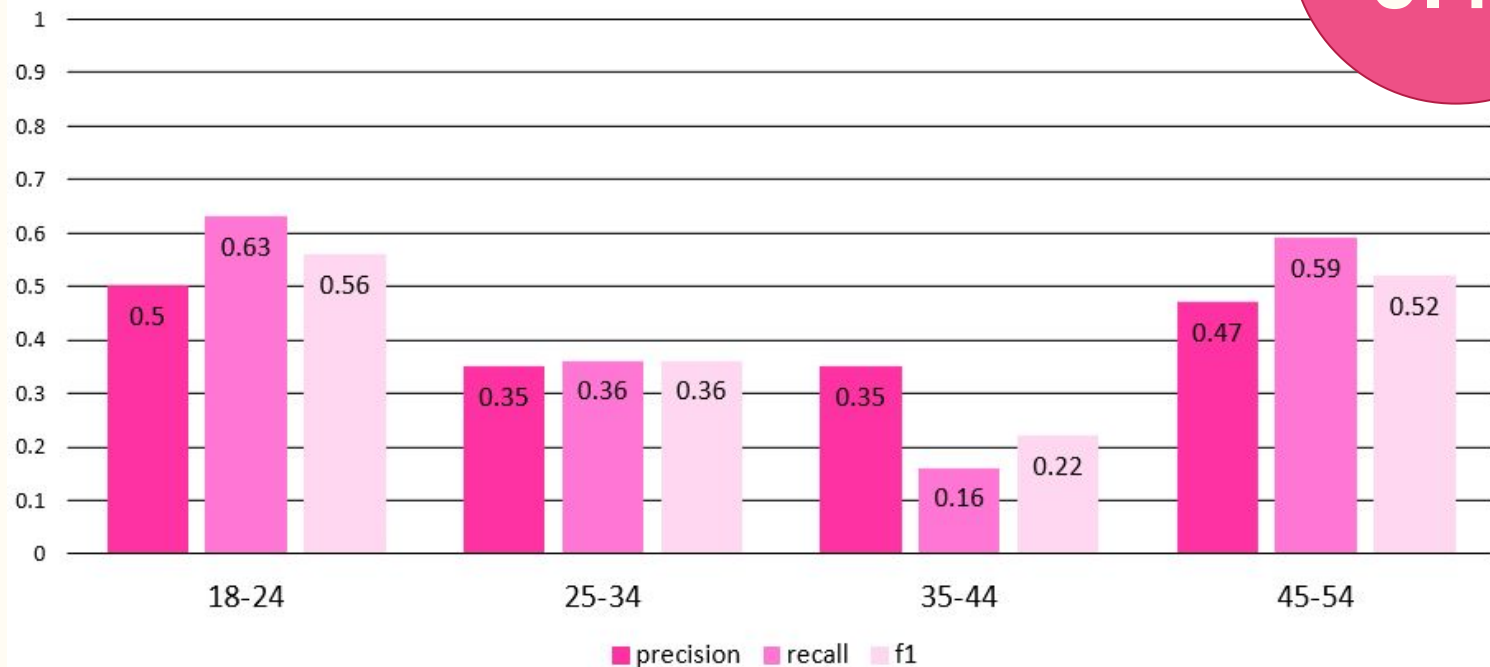


Approach 5: Results

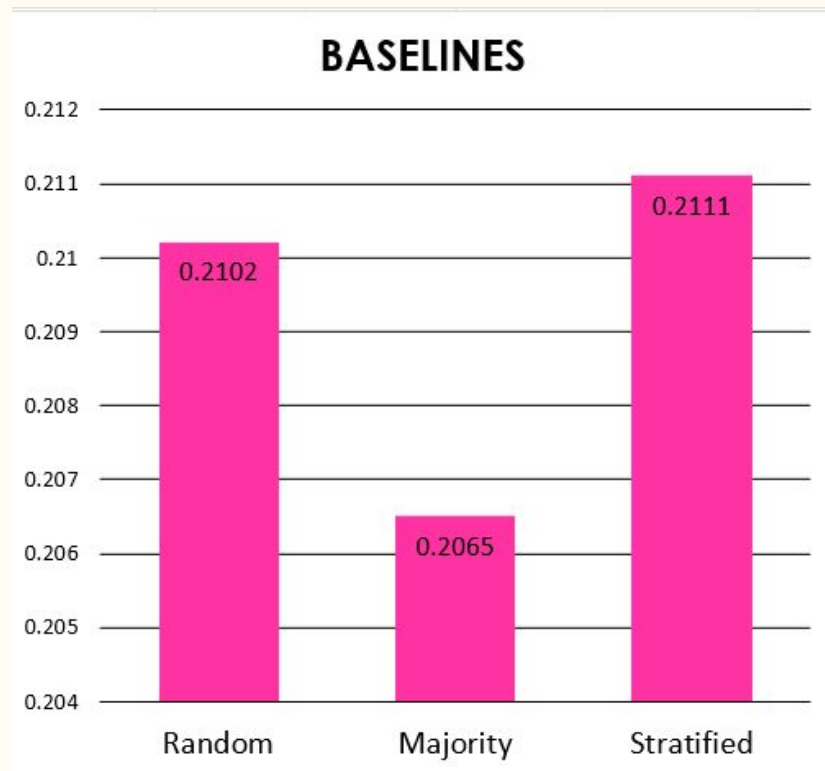
Accuracy:

0.43

BERT RESULTS



Baselines Results



Models Results



Analysis

- ★ Word2Vec, Td-idf, Mistral zero-shot, Mistral few-shot and BERT **outperformed** baselines
 - Almost doubled accuracy of the three baselines
- ★ Bag of Words model performed the **same** as baselines
 - Baselines and BoW had accuracy values of around 0.25
- ★ **Tf-idf** performed the **best**
 - Td-idf: Statistical approach
 - Mistral Few Shot, Word2Vec and BERT were slightly worse but comparable
- ★ Group 1, with ages from 18-24, was most accurately identified across all models except BoW which highly identified group 2



Conclusions & Future Work

Conclusions

Goal: predict approximate age of user given their essay responses

Summary:

- 1 Tf-idf
- 2 Mistral (few-shot)
- 2 Word2Vec
- 2 BERT
- 3 Mistral (zero-shot)
- 4 Bag of Words

Meaning:

It is very difficult to predict someone's age based off of short essay responses! This is likely due to the fact that language-based age classification is more intuitive in humans and harder for a computer to predict.

However, Tf-idf, a statistical approach performs the best.

Future Work

- ★ This model can be used for other datasets like identifying age from tweets, emails, etc.
 - Identify age of someone tweeting hate tweets
 - Identify which age groups use social media the most
- ★ Use for transfer learning with other social media apps to conduct further research
 - OkCupid is a small fraction of the population on social media
 - Much more popular dating apps now like Hinge and Bumble
- ★ For more accurate results, could try partitioning by gender



Thank you!

Github repo: <https://github.com/trailorr/NLP-Date-a-Base>